Characterizing Datasets: Hardness, Learning, and More

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July 8, 2026

Abstract

We develop elementary Fourier analysis *over a finite dataset* $\mathcal{D} \subset \mathcal{T}^n$: coefficients are taken with respect to the empirical measure on $\mathcal{D}$ while the characters remain those of the ambient product space, so that every quantity is estimable from samples. Parseval fails over a dataset — by exactly the density factor $C_{\mathcal{D}} = |\mathcal{T}|^n \,/\, |\mathcal{D}|$ — and we give the correct replacements: a mass identity, a closeness identity, and a low-degree learning theorem in which the normalization constants cancel through a scaled reconstruction. As the highlight, we adapt the Goldreich-Levin algorithm to find all characters heavily correlated with a function *over the dataset*. Sample access alone is provably blind: over a collision-free dataset, every bucket weight in the classical tree search is identical. Query access to a model together with the dataset's heavy bias spectrum suffices, via a convolution identity expressing the on-dataset spectrum as the global spectrum convolved with the spectrum of the dataset itself. Corollaries include a Kushilevitz-Mansour-style guarantee for decision trees over datasets.

1 Fourier Analysis on a Dataset

The classical theory of Boolean functions — and, more generally, Fourier analysis over product probability spaces — is one of the most versatile toolkits in theoretical computer science. Its basic premise, however, is that expectations are taken over a *product measure*: every coordinate is resampled independently. Real data is not like this. A corpus of token sequences, a folder of images, a table of feature vectors — each is a finite set $\mathcal{D} \subset \mathcal{T}^n$ inside an ambient product space, and the functions we care about (a trained model, a labeling rule) are only meaningfully evaluated *on the data*.

This paper develops the elementary theory that results from one stubborn design decision: **keep the characters of the ambient product space, but take all expectations over the dataset.** The resulting "dataset Fourier coefficients" are easy to estimate from samples and retain much of the classical dictionary — but not all of it, and the failures are precise, quantifiable, and (as we will see in the Goldreich-Levin section) algorithmically meaningful. Throughout, think of f as either a trained model, an ideal labeling function on the dataset, or some other efficiently computable function.

1.1 Orthonormal Bases and Parseval

We first fix the ambient framework, which is standard [1].

Definition 1.1 *(Probability Space with Orthonormal Basis)*: Let Ω be a measurable space with probability measure μ, and let I be an index set. An *orthonormal basis* for $L^2(\Omega, \mu)$ is a collection $\{\varphi_\alpha\}_{\alpha\in I}$ of functions $\varphi_\alpha : \Omega \to \mathbb{R}$ satisfying:
1. **Orthonormality:** $\langle \varphi_\alpha, \varphi_\beta \rangle_\mu = \mathbb{E}_{x\sim\mu}[\varphi_\alpha(x)\varphi_\beta(x)] = \delta_{\alpha\beta}$;
2. **Completeness:** every $f \in L^2(\Omega, \mu)$ can be written as $f = \sum_{\alpha\in I} \hat{f}(\alpha)\varphi_\alpha$.

We always take $\varphi_0 = 1$. For product spaces Ω^n with product measure $\mu^{\otimes n}$, the products $\varphi_\alpha = \prod_i \varphi_{\alpha_i}$ (over multi-indices α) form an orthonormal basis for $L^2(\Omega^n, \mu^{\otimes n})$.

Definition 1.2 *(Fourier Coefficients)*: For $f \in L^2(\Omega^n, \mu)$, the *Fourier coefficient* at α is

$$\hat{f}(\alpha) = \langle f, \varphi_\alpha \rangle_\mu = \mathbb{E}_{x\sim\mu}[f(x)\varphi_\alpha(x)],$$

so that $f = \sum_\alpha \hat{f}(\alpha)\varphi_\alpha$ (in L^2).

Proposition 1.1 *(Parseval)*: For any $f, g \in L^2(\Omega^n, \mu)$,

$$\langle f, g \rangle_\mu = \sum_\alpha \hat{f}(\alpha)\hat{g}(\alpha), \quad \text{in particular} \quad \mathbb{E}_{x\sim\mu}[f(x)^2] = \sum_\alpha \hat{f}(\alpha)^2.$$

Proof: Expand both functions in the basis and use orthonormality; the interchange of sum and integral is justified by L^2 convergence. ■

The two running examples to keep in mind:

Example: *(Haar / parity basis.)* $\Omega = \{-1, 1\}$ with the uniform measure. For $S \subseteq [n]$ the *parity functions* $\chi_S(x) = \prod_{i\in S} x_i$ form an orthonormal basis of $L^2(\{-1,1\}^n)$: for $S \neq T$ pick $i \in S\Delta T$ and note $\mathbb{E}[\chi_{S\Delta T}] = \mathbb{E}[x_i] \cdot \mathbb{E}\left[\chi_{S\Delta T\setminus\{i\}}\right] = 0$.

Example: *(Hermite basis.)* $\Omega = \mathbb{R}$ with the standard Gaussian measure γ. The orthonormalized *probabilist's Hermite polynomials* $h_0 = 1$, $h_1(x) = x$, $h_2(x) = (x^2 - 1)/\sqrt{2}$, ... form an orthonormal basis; their products $H_\alpha(x) = \prod_i h_{\alpha_i}(x_i)$, $\alpha \in \mathbb{N}^n$, handle Gaussian space $L^2(\mathbb{R}^n, \gamma_n)$, with $\deg H_\alpha = |\alpha| = \sum_i \alpha_i$. (The Ornstein-Uhlenbeck noise operator and the invariance principle port over as well; we will not need them here.)

1.2 Coordinate Averaging and Sensitivity

For product spaces there is a standard way to measure how much f depends on a coordinate.

Definition 1.3 *(Coordinate Averaging and Sensitivity)*: For $i \in [n]$, the *averaging operator* is $E^i[f](x) = \mathbb{E}_{a\sim\mu_i}[f(x^{i\leftarrow a})]$, where $x^{i\leftarrow a}$ replaces the i-th coordinate of x by a. The *sensitivity of coordinate i* (often called *influence*) and the *total sensitivity* are

$$\mathrm{Sens}_i[f] = \mathbb{E}_{x\sim\mu}\left[\left(f(x) - E^i[f](x)\right)^2\right], \qquad \boldsymbol{S}[f] = \sum_{i=1}^n \mathrm{Sens}_i[f].$$

Proposition 1.2: E^i kills exactly the basis directions that use coordinate i: $E^i[\varphi_\alpha] = \varphi_\alpha$ if $\alpha_i = 0$ and $E^i[\varphi_\alpha] = 0$ otherwise. Consequently $\mathrm{Sens}_i[f] = \sum_{\alpha:\alpha_i\neq 0} \hat{f}(\alpha)^2$ and $\boldsymbol{S}[f] = \sum_\alpha \#\alpha \cdot \hat{f}(\alpha)^2$, where $\#\alpha = |\{i : \alpha_i \neq 0\}|$.

Proof: The first claim is orthonormality in the i-th factor ($\mathbb{E}_{a\sim\mu_i}\left[\varphi_{\alpha_i}(a)\right] = \left\langle \varphi_{\alpha_i}, \varphi_0 \right\rangle = \delta_{\alpha_i,0}$); the rest is Parseval applied to $f - E^i f = \sum_{\alpha:\alpha_i\neq 0} \hat{f}(\alpha)\varphi_\alpha$. ■

1.3 Datasets: the Basic Dictionary

Now for the object of study. Fix a finite alphabet $\mathcal{T}$ and — for this subsection through the end of the paper — let μ be the **uniform** measure on $\mathcal{T}$, with $\{\varphi_\alpha\}$ any orthonormal basis of $L^2(\mathcal{T}^n)$ containing $\varphi_0 = 1$. (We remark below on what survives for non-uniform μ.)

Definition 1.4 *(Dataset; Dataset Fourier Coefficient)*: A *dataset* is a finite set $\mathcal{D} \subset \mathcal{T}^n$; we write $x \sim \mathcal{D}$ for a uniformly random element of $\mathcal{D}$ and

$$\hat{f}_{\mathcal{D}}(\alpha) = \mathbb{E}_{x\sim\mathcal{D}}[f(x)\varphi_\alpha(x)] = \frac{1}{|\mathcal{D}|}\sum_{x\in\mathcal{D}} f(x)\varphi_\alpha(x)$$

for the *dataset Fourier coefficient* of $f : \mathcal{D} \to \mathbb{R}$ at α. The *density constant* is $C_{\mathcal{D}} = |\mathcal{T}|^n \,/\, |\mathcal{D}|$.

Dataset coefficients are exactly the quantities estimable from labeled samples $(x, f(x))$, $x \sim \mathcal{D}$ — this is the point of the definition. The price: $\{\varphi_\alpha\}$ is orthonormal in $L^2(\mathcal{T}^n, \mu)$, *not* in $L^2(\mathcal{D})$. Two characters may even coincide as functions on $\mathcal{D}$ (we call this *aliasing*; the subcube example in the next section makes it vivid). So none of the Parseval-based dictionary may be borrowed blindly. The correct dictionary is routed through one object:

Definition 1.5 *(The Lift)*: For $f : \mathcal{D} \to \mathbb{R}$, the *lift* $\mathcal{D} \circ f : \mathcal{T}^n \to \mathbb{R}$ is

$$(\mathcal{D} \circ f)(x) = \begin{cases} f(x) & \text{if } x \in \mathcal{D} \\ 0 & \text{otherwise.} \end{cases}$$

Lemma 1.1 *(Lifting Lemma)*: $(\widehat{\mathcal{D} \circ f})(\alpha) = C_{\mathcal{D}}^{-1} \hat{f}_{\mathcal{D}}(\alpha)$ for every α.

Proof: $\mathbb{E}_{x\sim\mu}[(\mathcal{D} \circ f)(x)\varphi_\alpha(x)] = \frac{1}{|\mathcal{T}|^n}\sum_{x\in\mathcal{D}} f(x)\varphi_\alpha(x) = \frac{|\mathcal{D}|}{|\mathcal{T}|^n} \cdot \hat{f}_{\mathcal{D}}(\alpha)$. ∎

Everything true about datasets in this paper is the Lifting Lemma composed with an honest fact about $L^2(\mu)$. Three instances follow. First, inversion — the dataset coefficients determine f on $\mathcal{D}$:

Proposition 1.3 *(Inversion)*: For every $x \in \mathcal{D}$: $\quad f(x) = C_{\mathcal{D}}^{-1} \sum_\alpha \hat{f}_{\mathcal{D}}(\alpha)\varphi_\alpha(x)$.

Proof: Expand the lift in $L^2(\mu)$ and evaluate at $x \in \mathcal{D}$, where $(\mathcal{D} \circ f)(x) = f(x)$; apply Lemma 1.1. ∎

Second, the replacement for Parseval. There is no Parseval identity over $\mathcal{D}$; what is true is off by exactly the density constant.

Theorem 1.1 (Mass Identity — "Parseval over a dataset"): For any dataset $\mathcal{D}$ (distinct points) and any $f : \mathcal{D} \to \mathbb{R}$,

$$\sum_\alpha \hat{f}_{\mathcal{D}}(\alpha)^2 = C_{\mathcal{D}} \cdot \mathbb{E}_{x\sim\mathcal{D}}[f(x)^2].$$

Proof: Since $\{\varphi_\alpha\}$ is a complete orthonormal system for the uniform measure on the finite set $\mathcal{T}^n$, it satisfies the *reproducing kernel identity*

$$\sum_\alpha \varphi_\alpha(x)\varphi_\alpha(x') = |\mathcal{T}|^n \cdot \mathbb{1}[x = x']$$

(the matrix $M_{\alpha,x} = \varphi_\alpha(x)/\sqrt{|\mathcal{T}|^n}$ has orthonormal rows, hence orthonormal columns). Now expand and swap sums:

$$\sum_{\alpha} \hat{f}_{\mathcal{D}}(\alpha)^2 = \frac{1}{|\mathcal{D}|^2} \sum_{x,x' \in \mathcal{D}} f(x) f(x') \sum_{\alpha} \varphi_\alpha(x) \varphi_\alpha(x') = \frac{|\mathcal{T}|^n}{|\mathcal{D}|^2} \sum_{x \in \mathcal{D}} f(x)^2 = C_{\mathcal{D}} \cdot \mathbb{E}_{\mathcal{D}}[f^2].$$

■

For a non-uniform finite base measure the same proof gives the weighted form $\sum_\alpha \hat{f}_{\mathcal{D}}(\alpha)^2 = \frac{1}{|\mathcal{D}|^2} \sum_{x \in \mathcal{D}} f(x)^2/\mu(x)$; for continuous Ω (e.g., Gaussian space) the left side diverges — the empirical measure of a finite dataset has no $L^2(\mu)$ density. This is why we fix a finite alphabet.

Since $C_{\mathcal{D}}$ is enormous for small datasets, the total dataset Fourier mass is *not* $\mathbb{E}_{\mathcal{D}}[f^2]$; a function of unit norm on the dataset carries $C_{\mathcal{D}}$ worth of squared coefficients, smeared across aliases. Applying Theorem 1.1 to a difference gives the identity we will actually use:

Lemma 1.2 *(Parseval for Dataset Closeness)*: For $f, g : \mathcal{D} \to \mathbb{R}$,

$$\mathbb{E}_{x \sim \mathcal{D}}[(f(x) - g(x))^2] = C_{\mathcal{D}}^{-1} \sum_{\alpha} \left(\hat{f}_{\mathcal{D}}(\alpha) - \hat{g}_{\mathcal{D}}(\alpha)\right)^2.$$

Proof: By linearity of $\mathbb{E}_{\mathcal{D}}$, $(\widehat{f-g})_{\mathcal{D}}(\alpha) = \hat{f}_{\mathcal{D}}(\alpha) - \hat{g}_{\mathcal{D}}(\alpha)$; apply Theorem 1.1 to $f - g$. ■

Definition 1.6 *($\epsilon_{\mathcal{D}}$-closeness; Normalized Weights)*: We say g is *$\epsilon_{\mathcal{D}}$-close* to f if $\mathbb{E}_{x \sim \mathcal{D}}[(f(x) - g(x))^2] \le \epsilon$ — equivalently, $C_{\mathcal{D}} \cdot \mathbb{E}_{x \sim \mu}[((\mathcal{D} \circ f)(x) - (\mathcal{D} \circ g)(x))^2] \le \epsilon$. The *normalized level weights* are

$$\overline{W}^k_{\mathcal{D}}[f] = C_{\mathcal{D}}^{-1} \sum_{\#\alpha = k} \hat{f}_{\mathcal{D}}(\alpha)^2, \quad \text{so that} \quad \sum_{k \ge 0} \overline{W}^k_{\mathcal{D}}[f] = \mathbb{E}_{\mathcal{D}}[f^2]$$

by Theorem 1.1. Closeness is always measured *over the dataset*; only the coefficient side of Lemma 1.2 carries the normalization.

1.4 Averaging and Sensitivity on a Dataset

The third instance of the lift-then-use-$L^2(\mu)$ recipe concerns coordinate averaging. On a dataset, resampling a coordinate may leave the data; the natural operator zeros out such settings — and this choice makes it agree exactly with the global operator applied to the lift.

Definition 1.7 *(Dataset Coordinate Averaging; Dataset Sensitivity)*: For $x \in \mathcal{D}$,

$$E^i_{\mathcal{D}}[f](x) = \mathbb{E}_{a \sim \mu_i}[\mathcal{D}(x^{i \leftarrow a}) \cdot f(x^{i \leftarrow a})], \qquad \mathrm{Sens}^i_{\mathcal{D}}[f] = \mathbb{E}_{x \sim \mathcal{D}}\left[\left(f(x) - E^i_{\mathcal{D}}[f](x)\right)^2\right],$$

where $\mathcal{D}(\cdot)$ denotes the indicator of the dataset. Note $E^i_{\mathcal{D}}[f] = E^i[\mathcal{D} \circ f]$ restricted to $\mathcal{D}$, directly from the definitions.

Proposition 1.4 *(Dataset Sensitivity Bound)*:

$$\mathrm{Sens}^i_{\mathcal{D}}[f] \le C_{\mathcal{D}} \cdot \mathbb{E}_{x \sim \mu}\left[\left((\mathcal{D} \circ f) - E^i[\mathcal{D} \circ f]\right)^2\right] = C_{\mathcal{D}}^{-1} \sum_{\alpha : \alpha_i \ne 0} \hat{f}_{\mathcal{D}}(\alpha)^2,$$

and the inequality is an equality when $(\mathcal{D} \circ f) - E^i[\mathcal{D} \circ f]$ is supported on $\mathcal{D}$. Consequently $\boldsymbol{S}_{\mathcal{D}}[f] \coloneqq \sum_i \mathrm{Sens}^i_{\mathcal{D}}[f] \le \sum_{k \ge 1} k \cdot \overline{W}^k_{\mathcal{D}}[f]$.

Proof: Write $q = (\mathcal{D} \circ f) - E^i[\mathcal{D} \circ f]$, so that $\mathrm{Sens}^i_{\mathcal{D}}[f] = \mathbb{E}_{\mathcal{D}}[q^2] = C_{\mathcal{D}} \cdot \mathbb{E}_\mu[\mathcal{D} \cdot q^2] \le C_{\mathcal{D}} \cdot \mathbb{E}_\mu[q^2]$. Then apply Proposition 1.2 to the lift and Lemma 1.1; summing over i counts each α once per nonzero coordinate. ■

The inequality can be strict, and the naive analogue of Proposition 1.2 with dataset coefficients (no $C_{\mathcal{D}}^{-1}$) is simply false. A useful example to remember: $\mathcal{D} = \{x : x_1 = 1\} \subset \{-1, 1\}^n$ and $f =$

x_2. Then $\mathrm{Sens}^2_{\mathcal{D}}[f] = 1$, while $\chi_{\{2\}}$ and $\chi_{\{1,2\}}$ alias on $\mathcal{D}$, so $\sum_{\alpha:\alpha_2\neq 0} \hat{f}_{\mathcal{D}}(\alpha)^2 = 1 + 1 = 2$; the bound of Proposition 1.4 gives $2/C_{\mathcal{D}} = 1$ (here $C_{\mathcal{D}} = 2$) — tight.

1.5 Learning Low-Degree Functions from Dataset Samples

As a first payoff, the classical Low-Degree Algorithm [1] survives on datasets, *provided* the reconstruction is scaled by $C_{\mathcal{D}}^{-1}$. The scaling matters: in the half-cube example above, the unscaled plug-in reconstruction of $f = x_2$ at degree 2 returns $\chi_2 + \chi_{12} = 2x_2$ on $\mathcal{D}$ — error 1 with exact coefficients and zero tail — because each character is counted once per alias. The scaled reconstruction returns $(x_2 + x_1x_2)/2 = x_2$ on $\mathcal{D}$, exactly. In general, the $C_{\mathcal{D}}$ of Lemma 1.2 cancels against the two factors of $C_{\mathcal{D}}^{-1}$ in the scaled output, via Proposition 1.3.

Theorem 1.2 (Learning Low-Degree Functions over a Dataset): Let $f : \mathcal{D} \to [-1, 1]$ and suppose the normalized high-degree weight satisfies

$$\sum_{k>d} \overline{W}^k_{\mathcal{D}}[f] \leq \frac{\epsilon^2}{4}$$

(e.g., $d \geq 4\overline{S}/\epsilon^2$ suffices when $\sum_k k \cdot \overline{W}^k_{\mathcal{D}}[f] \leq \overline{S}$). Given $m = O\left(\frac{n^d}{\epsilon^2} \cdot \log\left(n^d/\delta\right)\right)$ samples from $\mathcal{D}$, one can output h with $\mathbb{E}_{x\sim\mathcal{D}}\left[(f(x) - h(x))^2\right] \leq \epsilon^2$ with probability at least $1 - \delta$.

Proof: **Algorithm.** For each $\#\alpha \leq d$, estimate $\hat{f}_{\mathcal{D}}(\alpha)$ by $\tilde{f}(\alpha) = \frac{1}{m}\sum_{j=1}^m f\left(x^{(j)}\right)\varphi_\alpha\left(x^{(j)}\right)$; output the *scaled* reconstruction $h = C_{\mathcal{D}}^{-1} \sum_{\#\alpha\leq d} \tilde{f}(\alpha)\varphi_\alpha$.

Analysis. Write $g^\uparrow = \mathcal{D} \circ f$, so $\hat{g^\uparrow}(\alpha) = C_{\mathcal{D}}^{-1}\hat{f}_{\mathcal{D}}(\alpha)$ (Lemma 1.1) and $f = g^\uparrow$ on $\mathcal{D}$. Dropping the indicator and applying Parseval in $L^2(\mu)$:

$$\begin{aligned}\mathbb{E}_{\mathcal{D}}\left[(f-h)^2\right] = C_{\mathcal{D}} \cdot \mathbb{E}_\mu\left[\mathcal{D} \cdot \left(g^\uparrow - h\right)^2\right] &\leq C_{\mathcal{D}} \cdot \mathbb{E}_\mu\left[\left(g^\uparrow - h\right)^2\right] \\ &= C_{\mathcal{D}} \sum_\alpha \left(\hat{g^\uparrow}(\alpha) - \hat{h}(\alpha)\right)^2.\end{aligned}$$

Since $\hat{h}(\alpha) = C_{\mathcal{D}}^{-1}\tilde{f}(\alpha)$ for $\#\alpha \leq d$ and 0 otherwise, the $C_{\mathcal{D}}$ cancels against the two factors of $C_{\mathcal{D}}^{-1}$:

$$\mathbb{E}_{\mathcal{D}}\left[(f-h)^2\right] \leq \underbrace{\sum_{k>d} \overline{W}^k_{\mathcal{D}}[f]}_{\leq \epsilon^2/4 \text{ (hypothesis)}} + \underbrace{C_{\mathcal{D}}^{-1} \sum_{\#\alpha\leq d} \left(\hat{f}_{\mathcal{D}}(\alpha) - \tilde{f}(\alpha)\right)^2}_{\leq \epsilon^2/4 \text{ w.h.p.}}.$$

For the second term: each $\tilde{f}(\alpha)$ averages m i.i.d. terms in $[-1, 1]$ with mean $\hat{f}_{\mathcal{D}}(\alpha)$; by Hoeffding and a union bound over the $O\left(n^d\right)$ low-degree indices, the stated m makes every deviation at most $\epsilon/\left(2n^{d/2}\right)$, so the sum is at most $\epsilon^2/4$ even before the (generous) factor $C_{\mathcal{D}}^{-1} \leq 1$. ■

Two remarks, in the spirit of honesty. First, the hypothesis is stated in normalized weights, the correct analogue of "ϵ^2 of Fourier weight above degree d"; in raw dataset units it reads $\sum_{k>d}\sum_{\#\alpha=k} \hat{f}_{\mathcal{D}}(\alpha)^2 \leq C_{\mathcal{D}}\epsilon^2/4$. Second, for a *generic* sparse dataset the hypothesis is genuinely demanding — it asks the lift $\mathcal{D} \circ f$ to be low-degree concentrated in $L^2(\mu)$ — and the next section shows this is not an artifact of our proof but a structural fact about datasets. Structured datasets (dense ones, or subcube-like ones as in the example above) satisfy it naturally.

2 Highlight: Goldreich-Levin over a Dataset

We now come to the highlight: adapting the Goldreich-Levin algorithm ([1], Section 3.5) to the dataset setting. The classical algorithm, given query access to f, finds *all* of the characters with which f is noticeably correlated — the engine behind the Kushilevitz-Mansour learning algorithm. Our use case is the on-distribution version of the same question: f is a trained model (or labeling rule), $\mathcal{D}$ is the data, and we want every character that is noticeably correlated with f *over the dataset*, i.e., every heavy dataset coefficient $\hat{f}_{\mathcal{D}}(S)$. These are exactly the coefficients that describe f's behavior where it is actually evaluated, and — when the dataset is anything but uniform — they differ from the global ones.

For clarity of exposition we work over the Boolean cube $\Omega^n = \{-1, 1\}^n$ with the parity characters χ_S, writing $\hat{f}_{\mathcal{D}}(S) = \mathbb{E}_{x\sim\mathcal{D}}[f(x)\chi_S(x)]$ as before.[1] The goal, mirroring the classical guarantee:

> **Definition 2.1** *(Dataset Heavy-Coefficient Search)*: Given $0 < \tau \leq 1$ and access to f and $\mathcal{D}$ (in a model to be specified), output a list L of subsets of $[n]$ such that with high probability:
> 1. (Completeness) $|\hat{f}_{\mathcal{D}}(S)| \geq \tau \implies S \in L$;
> 2. (Soundness) $S \in L \implies |\hat{f}_{\mathcal{D}}(S)| \geq \tau/2$.

2.1 Access Models

The choice of access model turns out to be the crux of the problem. We distinguish three oracles:
1. **Samples** (SAMP): i.i.d. draws $x \sim \mathcal{D}$ together with the label $f(x)$.
2. **Membership** (MEM): the in-distribution tester $x \mapsto \mathcal{D}(x) \in \{0, 1\}$, evaluable on *arbitrary* $x \in \{-1, 1\}^n$.
3. **Model queries** (QUERY): f evaluable at arbitrary $x \in \{-1, 1\}^n$, including out-of-distribution points — the natural model when f is a trained model or some other efficiently computable function.

Classical GL uses only QUERY, and its guarantee concerns the *global* coefficients $\hat{f}(S)$. For the *dataset* coefficients, we will see that samples alone are provably insufficient (Lemma 2.2), and that the positive results route all of the dataset's structure through a single object: its *bias spectrum* (Lemma 2.3).

[1] Every identity below that uses only orthonormality and the reproducing kernel $\sum_S \chi_S(x)\chi_S(x') = 2^n \cdot \mathbb{1}[x = x']$ extends verbatim to any finite abelian group with its character basis (with $\chi_S\overline{\chi_T}$ in place of $\chi_S\chi_T$), and to general product orthonormal bases over finite sets. The convolution identity additionally uses the group structure $\chi_S\chi_T = \chi_{S\Delta T}$.

2.2 Recap: the Classical Goldreich-Levin Algorithm

Theorem 2.1 (Goldreich-Levin; [1] Section 3.5): Given query access to $f : \{-1,1\}^n \to \{-1,1\}$ and $0 < \tau \leq 1$, there is a poly$(n, 1/\tau)$-time algorithm that with high probability outputs a list $L = \{U_1, ..., U_\ell\}$ of subsets of $[n]$ such that:
1. $|\hat{f}(U)| \geq \tau \Longrightarrow U \in L$;
2. $U \in L \Longrightarrow |\hat{f}(U)| \geq \tau/2$.

By Parseval, the second guarantee implies $|L| \leq 4/\tau^2$.

The engine is a divide-and-conquer search over the tree of coordinate prefixes. For $S \subseteq J \subseteq [n]$, define the **bucket weight** ([1], Definition 3.39)

$$W^{S \mid J}[f] = \sum_{T \subseteq \overline{J}} \hat{f}(S \cup T)^2,$$

the Fourier weight of f on sets whose restriction to J equals S. The algorithm maintains the buckets of weight at least $\tau^2/2$ — at most $4/\tau^2$ of them, by Parseval — splitting one coordinate at a time until each survivor is a singleton $\{S\}$ with $\hat{f}(S)^2 \geq \tau^2/2$. What makes this feasible is that bucket weights are *estimable*: by the restriction identity ([1], equation (3.5) and Proposition 3.40),

$$W^{S \mid J}[f] = \mathbb{E}_{z \sim \{-1,1\}^{\overline{J}}}\left[\hat{f}_{J \mid z}(S)^2\right] = \mathbb{E}_{z \sim \{-1,1\}^{\overline{J}}} \mathbb{E}_{y, y' \sim \{-1,1\}^J}[f(y,z)\chi_S(y) \cdot f(y',z)\chi_S(y')],$$

so a QUERY algorithm gets accuracy $\pm\epsilon$ with $O(\log(1/\delta)/\epsilon^2)$ queries.

Two remarks for later use. First, the proof only needs $\|f\|_\infty \leq 1$ (for the Chernoff estimates) and $\|f\|_2 \leq 1$ (for the Parseval pruning bound), so Theorem 2.1 holds verbatim for $f : \{-1,1\}^n \to [-1,1]$, with list size $4\|f\|_2^2/\tau^2$. Second, the running time is polynomial in $1/\tau$ — the quantity that will blow up if we naively lift a sparse dataset.

2.3 Obstructions

Why can't we just run the same tree search on the dataset coefficients? Recall from the Mass Identity (Theorem 1.1) that $\sum_S \hat{f}_{\mathcal{D}}(S)^2 = C_{\mathcal{D}} \cdot \mathbb{E}_{\mathcal{D}}[f^2]$: Parseval fails over $\mathcal{D}$ by exactly the density factor. This already kills the pruning bound — the number of τ-heavy dataset coefficients can be as large as $C_{\mathcal{D}}\mathbb{E}_{\mathcal{D}}[f^2]/\tau^2$, i.e., exponential — and the failure is not confined to exotic functions:

Example: Let $K \subseteq [n]$ and let $\mathcal{D} = \{x : x_i = 1 \text{ for all } i \in K\}$ be the subcube fixing K, with $f \equiv 1$. Every χ_S with $S \subseteq K$ is identically 1 on $\mathcal{D}$, so

$$\hat{f}_{\mathcal{D}}(S) = \begin{cases} 1 \text{ if } S \subseteq K \\ 0 \text{ otherwise,} \end{cases}$$

and there are $2^{|K|}$ maximal coefficients: distinct characters are *identical as functions on* $\mathcal{D}$, and "the" heavy set is only defined up to this aliasing.

More damning still: the bucket weights themselves carry *no information* over a generic dataset. The following exact identity is the dataset analogue of the restriction identity.

Lemma 2.1 *(Pair Form of Dataset Bucket Weights)*: Fix $J \subseteq [n]$ with $|J| = k$ and $S \subseteq J$. Then

$$\sum_{U \subseteq \overline{J}} \hat{f}_{\mathcal{D}}(S \cup U)^2 = 2^{n-k} \cdot \mathbb{E}_{x,x' \sim \mathcal{D}}[f(x)f(x')\chi_S(x_J)\chi_S(x'_J) \cdot \mathbb{1}[x_{\overline{J}} = x'_{\overline{J}}]].$$

Proof: Expand each $\hat{f}_{\mathcal{D}}(S \cup U) = \mathbb{E}_{x\sim\mathcal{D}}[f(x)\chi_S(x_J)\chi_U(x_{\overline{J}})]$, square, and swap sums; the inner sum $\sum_{U\subseteq\overline{J}} \chi_U(x_{\overline{J}})\chi_U(x'_{\overline{J}})$ is the reproducing kernel $2^{n-k} \cdot \mathbb{1}[x_{\overline{J}} = x'_{\overline{J}}]$ on $\{-1,1\}^{\overline{J}}$.■

Lemma 2.2 *(Blindness of Dataset Bucket Weights)*: In the setting of Lemma 2.1, suppose no two dataset points collide on $\overline{J}$ (the projection $x \mapsto x_{\overline{J}}$ is injective on $\mathcal{D}$). Then for *every* $S \subseteq J$,

$$\sum_{U \subseteq \overline{J}} \hat{f}_{\mathcal{D}}(S \cup U)^2 = \frac{2^{n-k}}{|\mathcal{D}|} \cdot \mathbb{E}_{x\sim\mathcal{D}}[f(x)^2],$$

independent of S.

Proof: Under injectivity only the diagonal of Lemma 2.1 survives, and $\chi_S(x_J)^2 = 1$. ■

In words: as long as the dataset's projection onto the un-split coordinates is collision-free, every bucket at level k has exactly the same weight — no matter what f is, and no matter where its heavy dataset coefficients lie. For m points in general position, a union bound gives collisions on $\overline{J}$ with probability at most $\binom{m}{2}/2^{n-k}$, so the search tree is information-free for all levels $k \lesssim n - 2\log_2 m$ — the *birthday horizon.* A GL-style search would descend blindly through $2^{n-2\log_2 m}$ buckets before the weights begin to distinguish anything. Together with the subcube example, this rules out a sample-only dataset GL: additional access to f, or structural knowledge of $\mathcal{D}$, is necessary.[2]

2.4 The Convolution Identity

The positive results all flow from one structural fact: the dataset spectrum of f is the global spectrum of f *convolved with the spectrum of the dataset itself.*

Definition 2.2 *(Bias Spectrum)*: The **bias spectrum** of $\mathcal{D}$ is the family

$$b_V = \mathbb{E}_{x\sim\mathcal{D}}[\chi_V(x)], \quad V \subseteq [n]$$

— the dataset Fourier coefficients of the constant function 1. Note $b_\emptyset = 1$ and $|b_V| \leq 1$. For $\theta > 0$, write $B_\theta = \{V : |b_V| \geq \theta\}$ for the **$\boldsymbol{\theta}$-heavy bias set**.

Lemma 2.3 *(Convolution Identity)*: For any $f : \{-1,1\}^n \to \mathbb{R}$ and any $S \subseteq [n]$,

$$\hat{f}_{\mathcal{D}}(S) = \sum_{V \subseteq [n]} b_V \cdot \hat{f}(S \Delta V),$$

where Δ denotes symmetric difference.

Proof: Expand f in its global Fourier expansion inside the dataset expectation:

$$\hat{f}_{\mathcal{D}}(S) = \sum_T \hat{f}(T) \cdot \mathbb{E}_{x\sim\mathcal{D}}[\chi_T(x)\chi_S(x)] = \sum_T \hat{f}(T) \cdot b_{T\Delta S},$$

using $\chi_T\chi_S = \chi_{T\Delta S}$; substitute $V = T\Delta S$. ■

The bias spectrum is exactly the aliasing structure of the subcube example: there $b_V = \mathbb{1}[V \subseteq K]$, and the convolution smears the single global coefficient $\hat{f}(\emptyset) = 1$ onto all $2^{|K|}$ sets $S \subseteq K$. The blindness of Lemma 2.2 is the statement that for a generic dataset the bias spectrum is an

[2] A sanity check for Lemma 2.2: $n = 2$, $\mathcal{D} = \{(1,1), (-1,-1)\}$, $f(x) = x_1$, $J = \{1\}$. Then $\hat{f}_{\mathcal{D}}(\{1\}) = 1$, $\hat{f}_{\mathcal{D}}(\{1,2\}) = 0$, and the bucket weight is $1 = \frac{2^{2-1}}{2} \cdot 1$: the signal coefficient and the aliasing noise conspire to a flat bucket profile. All identities and theorem constants in this section have been verified numerically on random and structured instances.

exponentially wide field of $\sim \pm 1/\sqrt{|\mathcal{D}|}$ noise, which buries any bucket-level signal. Conversely, whenever the bias spectrum is *dominated by few heavy terms*, the dataset spectrum of f is a controlled deformation of its global spectrum — precisely what the main theorem exploits.

2.5 Main Theorem: Dataset GL from Model Queries

The standing assumptions: f is a queryable model of bounded *spectral norm* $\|\hat{f}\|_1 = \sum_T |\hat{f}(T)|$, and we know the heavy part of the dataset's bias spectrum. The latter is a structural input about $\mathcal{D}$ alone, not about f; we discuss how to obtain it in the next subsection.

Theorem 2.2 (Dataset Goldreich-Levin, model-query access): Let $f : \{-1,1\}^n \to [-1,1]$ with $\|\hat{f}\|_1 \le A$ be given by QUERY access, and let $\mathcal{D}$ be given by SAMP access. Let $0 < \tau \le 1$, $\delta \in (0,1)$, fix any $\theta \le \tau/(4A)$, and suppose the θ-heavy bias set B_θ of $\mathcal{D}$ is given explicitly. Then there is an algorithm running in time $\mathrm{poly}(n, |B_\theta|, 1/\tau, \log(1/\delta))$, using $\mathrm{poly}(n, |B_\theta|, 1/\tau) \cdot \log(1/\delta)$ queries to f and $O(\log(|B_\theta| / (\tau\delta))/\tau^2)$ samples from $\mathcal{D}$, that with probability at least $1-\delta$ outputs a list L satisfying:
1. (Completeness) $|\hat{f}_{\mathcal{D}}(S)| \ge \tau \Longrightarrow S \in L$;
2. (Soundness) $S \in L \Longrightarrow |\hat{f}_{\mathcal{D}}(S)| \ge \tau/2$;
3. $|L| \le \frac{64\,|B_\theta|^3}{9\tau^2}$.

The algorithm:
1. Run classical GL (Theorem 2.1, in its $[-1,1]$-valued form) on f with threshold $\tau' = \frac{3\tau}{4\,|B_\theta|}$ and confidence $1-\delta/2$, obtaining a list F of global heavy sets, $|F| \le 4/\tau'^2$.
2. Form the candidate list $C = \{T \Delta V : T \in F, V \in B_\theta\}$, so $|C| \le |F| \cdot |B_\theta|$.
3. Draw $m = O(\log(|C| / \delta)/\tau^2)$ samples from $\mathcal{D}$ and, for each $S \in C$, compute the empirical coefficient $\tilde{f}_{\mathcal{D}}(S) = \frac{1}{m}\sum_{j=1}^m f(x^{(j)})\chi_S(x^{(j)})$. Output $L = \left\{S \in C : |\tilde{f}_{\mathcal{D}}(S)| \ge 3\tau/4\right\}$.

Proof: **Decomposition.** By Lemma 2.3, for every S,

$$\hat{f}_{\mathcal{D}}(S) = \sum_{V \in B_\theta} b_V \hat{f}(S \Delta V) + R_S, \quad \text{where } |R_S| \le \sum_{V \notin B_\theta} |b_V|\ |\hat{f}(S\Delta V)| \le \theta \cdot \|\hat{f}\|_1 \le \theta A \le \frac{\tau}{4},$$

using $|b_V| < \theta$ off the heavy set and re-indexing the tail sum over $T = S\Delta V$.

Completeness. Suppose $|\hat{f}_{\mathcal{D}}(S)| \ge \tau$. Then

$$\sum_{V \in B_\theta} |b_V|\ |\hat{f}(S\Delta V)| \ge |\hat{f}_{\mathcal{D}}(S)| - |R_S| \ge \tau - \frac{\tau}{4} = \frac{3\tau}{4},$$

and since $|b_V| \le 1$, some $V^* \in B_\theta$ has $|\hat{f}(S\Delta V^*)| \ge \frac{3\tau}{4\,|B_\theta|} = \tau'$. By the completeness of classical GL (except with probability $\delta/2$), $S\Delta V^* \in F$, hence $S = (S\Delta V^*)\Delta V^* \in C$. By Hoeffding and a union bound over C (except with probability $\delta/2$), every empirical coefficient satisfies $|\tilde{f}_{\mathcal{D}}(S) - \hat{f}_{\mathcal{D}}(S)| \le \tau/8$; in particular $|\tilde{f}_{\mathcal{D}}(S)| \ge \tau - \tau/8 > 3\tau/4$, so $S \in L$.

Soundness. If $S \in L$ then $|\hat{f}_{\mathcal{D}}(S)| \ge 3\tau/4 - \tau/8 = 5\tau/8 \ge \tau/2$.

List size and complexity. $|L| \le |C| \le |F|\ |B_\theta| \le \frac{4}{\tau'^2}\ |B_\theta| = \frac{64\,|B_\theta|^3}{9\tau^2}$. Classical GL at threshold τ' costs $\mathrm{poly}(n, 1/\tau') = \mathrm{poly}(n, |B_\theta|, 1/\tau)$ queries; step 3 costs m samples and $m \cdot |C|$ arithmetic operations. ■

Note that the algorithm uses only the *set* B_θ, never the values b_V; and the theorem is meaningful precisely in the distribution-shift regime where $\hat{f}_{\mathcal{D}}(S)$ and $\hat{f}(S)$ differ — the shift is routed through the dataset's heavy biases. Running the completeness argument purely combinatorially (Parseval bounds the number of τ'-heavy global coefficients by $1/\tau'^2$) also yields an unconditional structural bound:

Corollary 2.1 *(List-Size Bound under Sparse Bias Spectrum)*: If $\left\|\hat{f}\right\|_1 \leq A$ and $\theta \leq \tau/(4A)$, then

$$|\left\{S : |\hat{f}_{\mathcal{D}}(S)| \geq \tau\right\}| \leq \frac{16\,|B_\theta|^3}{9\tau^2}.$$

2.6 Companion Results

Dense Datasets: the Membership-Oracle Route

When the dataset is *dense* — $C_{\mathcal{D}} = \mathrm{poly}(n)$ — the naive lifting strategy already works, and moreover supplies the heavy bias set B_θ needed above. Recall the lift $g = \mathcal{D} \circ f$, evaluable anywhere given MEM together with the ability to evaluate f on points of $\mathcal{D}$; its global coefficients satisfy $\hat{g}(S) = C_{\mathcal{D}}^{-1} \hat{f}_{\mathcal{D}}(S)$ (Lemma 1.1).

Proposition 2.1 *(Dataset GL for Dense Datasets)*: Given query access to $g = \mathcal{D} \circ f$ with $\|f\|_\infty \leq 1$, and $0 < \tau \leq 1$, running classical GL on g with threshold $C_{\mathcal{D}}^{-1}\tau$ solves the search problem of Definition 2.1 in time $\mathrm{poly}(n, C_{\mathcal{D}}, 1/\tau)$, with list size $|L| \leq 4C_{\mathcal{D}}\mathbb{E}_{\mathcal{D}}[f^2]/\tau^2$.

Proof: g takes values in $[-1,1]$, so the $[-1,1]$-valued form of Theorem 2.1 applies at threshold $\tau'' = C_{\mathcal{D}}^{-1}\tau$: completeness and soundness transfer through $\hat{g}(S) = C_{\mathcal{D}}^{-1}\hat{f}_{\mathcal{D}}(S)$, the running time is $\mathrm{poly}(n, 1/\tau'') = \mathrm{poly}(n, C_{\mathcal{D}}, 1/\tau)$, and Parseval gives $|L| \leq 4\|g\|_2^2/\tau''^2 = 4C_{\mathcal{D}}^{-1}\mathbb{E}_{\mathcal{D}}[f^2] \cdot C_{\mathcal{D}}^2/\tau^2$. ■

The list-size bound matches the Mass Identity (Theorem 1.1) exactly, so Proposition 2.1 is tight in general; and by taking $f \equiv 1$, the same run recovers the heavy bias set — $b_V = C_{\mathcal{D}} \cdot \hat{\mathbb{1}}_{\mathcal{D}}(V)$ — so B_θ is computable in time $\mathrm{poly}(n, C_{\mathcal{D}}, 1/\theta)$ given MEM. For sparse datasets ($C_{\mathcal{D}}$ superpolynomial) this route is unavailable, consistent with Lemma 2.2, and B_θ must come from prior structural knowledge of the dataset (known feature constraints, class balance, template structure).

Representative Datasets

At the opposite extreme, if the dataset's spectrum uniformly tracks the global one — the "no distribution shift" regime — dataset GL degenerates gracefully to classical GL plus re-scoring.

Definition 2.3 *(ϵ-Representative Dataset)*: $\mathcal{D}$ is **ϵ-representative** for f if $|\hat{f}_{\mathcal{D}}(S) - \hat{f}(S)| \leq \epsilon$ for all $S \subseteq [n]$.

Lemma 2.4: Let $f : \{-1,1\}^n \to [-1,1]$ and let $\mathcal{D}$ consist of m i.i.d. uniform samples. If $m \geq \frac{2}{\epsilon^2}\left(n \ln 2 + \ln \frac{2}{\delta}\right)$, then with probability at least $1-\delta$, $\mathcal{D}$ is ϵ-representative for f.

Proof: For fixed S, $\hat{f}_{\mathcal{D}}(S)$ averages m i.i.d. terms $f(x)\chi_S(x) \in [-1,1]$ with mean $\hat{f}(S)$; Hoeffding gives $\Pr\left[\left|\hat{f}_{\mathcal{D}}(S) - \hat{f}(S)\right| > \epsilon\right] \leq 2e^{-m\epsilon^2/2}$. Union bound over all 2^n sets. ■

Proposition 2.2 *(Dataset GL for Representative Datasets)*: Suppose $\mathcal{D}$ is $(\tau/8)$-representative for $f : \{-1,1\}^n \to [-1,1]$, and we have QUERY access to f and SAMP access to $\mathcal{D}$. Then the search problem of Definition 2.1 is solvable in time $\mathrm{poly}(n, 1/\tau, \log(1/\delta))$ with $O(\log(1/(\tau\delta))/\tau^2)$ samples: run classical GL at threshold $7\tau/8$

to get F, estimate $\hat{f}_{\mathcal{D}}(S)$ for $S \in F$ to accuracy $\tau/8$ from samples, and keep $|\tilde{f}_{\mathcal{D}}(S)| \geq 3\tau/4$.

Proof: If $|\hat{f}_{\mathcal{D}}(S)| \geq \tau$ then representativeness gives $|\hat{f}(S)| \geq \tau - \tau/8 = 7\tau/8$, so $S \in F$ by GL completeness; its estimate satisfies $|\tilde{f}_{\mathcal{D}}(S)| \geq |\hat{f}_{\mathcal{D}}(S)| - \tau/8 \geq 7\tau/8 > 3\tau/4$, so S is kept. (The representativeness slack is needed only for the GL step; the estimation step compares directly against $\hat{f}_{\mathcal{D}}(S)$.) If S is kept then $|\hat{f}_{\mathcal{D}}(S)| \geq 3\tau/4 - \tau/8 = 5\tau/8 \geq \tau/2$. GL's Parseval bound gives $|F| \leq 4/(7\tau/8)^2 \leq 7/\tau^2$, so a union bound over F needs $O(\log(|F|\ /\delta)/\tau^2)$ samples. ∎

In this regime the heavy dataset coefficients essentially *are* the heavy global coefficients; the substantive new content of Theorem 2.2 lies exactly where representativeness fails.

2.7 Application: Kushilevitz-Mansour over Datasets

Classical GL powers the Kushilevitz-Mansour learning algorithm ([1], Theorems 3.37-3.38): any f with $\left\|\hat{f}\right\|_1 \leq s$ — in particular any decision tree of size s — is learnable from queries in time $\mathrm{poly}(n, s, 1/\epsilon)$. The spectral-norm hypothesis is exactly the A of Theorem 2.2, so the on-distribution analogue of the coefficient-collection step comes for free:

Corollary 2.2 *(Heavy On-Dataset Coefficients of Decision Trees)*: Let $f : \{-1,1\}^n \to \{-1,1\}$ be computable by a decision tree of size s (so $\left\|\hat{f}\right\|_1 \leq s$), given by QUERY access, and let B_θ be the θ-heavy bias set of $\mathcal{D}$ for some $\theta \leq \tau/(4s)$. Then all S with $|\hat{f}_{\mathcal{D}}(S)| \geq \tau$ can be listed in time $\mathrm{poly}(n, s, |B_\theta|, 1/\tau, \log(1/\delta))$.

We emphasize what Corollary 2.2 does *not* yet give: an $\epsilon_{\mathcal{D}}$-close sparse approximation to f. Classically, coefficient collection plus Parseval yields the learning guarantee; over a dataset, the closeness identity carries a $C_{\mathcal{D}}^{-1}$ (Lemma 1.2), so bounding $\mathbb{E}_{x\sim\mathcal{D}}[(f-g)^2]$ for a sparse g assembled from the recovered coefficients requires genuinely new arguments. TODO: On-dataset approximation from recovered coefficients. One route: $\mathbb{E}_{\mathcal{D}}[(f-g)^2] = \sum_V b_V \cdot \widehat{(f-g)^2}(V)$ by Lemma 2.3 applied to $(f-g)^2$ at $S = \emptyset$, splitting the sum over B_θ and its complement; the tail is controlled by $\theta \cdot \left\|\widehat{(f-g)^2}\right\|_1 \leq \theta\left(\left\|\hat{f}\right\|_1 + \|\hat{g}\|_1\right)^2$, but the heavy-bias terms need per-V control. Compare with the $C_{\mathcal{D}}^{-1}$-scaled reconstruction route of Theorem 1.2.

2.8 Discussion and Open Directions

TODO:

- Non-uniform (weighted) datasets: the identities above should extend with $p(x)$ in place of $1/|\mathcal{D}|$; the reproducing-kernel steps are unaffected, the mass identity picks up $\|p\|_2^2$-type factors.
- General product alphabets and orthonormal bases (Hermite, tokens): the pair-form and blindness lemmas need only the reproducing kernel; the convolution identity needs the group structure ($\chi_S\overline{\chi_T} = \chi_{S-T}$), so state it over $\mathbb{Z}_q^n$ or general finite abelian groups.
- How to *certify* or *learn* the heavy bias set B_θ for structured real datasets (images, token corpora)? For dense datasets Proposition 2.1 computes it; for sparse ones, what natural structure (subcube constraints, sparsity templates, group symmetries) makes B_θ small and known?
- Relation to the "Active Fourier Auditor" of arXiv:2410.08111, which estimates distributional properties of ML models from queries and samples.

- Lower bound: turn Lemma 2.2 into a formal sample-complexity lower bound for Definition 2.1 in the SAMP-only model (two datasets, same visible statistics, different heavy sets).

3 Conclusion and Future Directions

Taking expectations over a dataset while keeping the characters of the ambient product space yields a usable, sample-estimable Fourier calculus — provided one respects the density constant $C_{\mathcal{D}}$ and the aliasing it encodes. The dictionary (lift, inversion, mass identity, closeness) is elementary; the payoff is that classical spectral algorithms can be transplanted honestly: low-degree learning survives with a scaled reconstruction, and Goldreich-Levin survives exactly when the dataset's bias spectrum is dominated by few heavy terms — with a matching impossibility (blindness) for sample-only access.

Beyond the open problems listed at the end of the Goldreich-Levin section, some directions we find promising:

TODO:

- *Degree monotonicity.* Is there a monotone relationship between the degree (spectral concentration) of f over the dataset and over the full space? A clean statement here would let one read off, from dataset quantities alone, whether a given model class can learn f — connect to distribution-agnostic learning theory.
- *The dataset as its own function.* What happens when $f = \mathcal{D}$ (the indicator itself)? The bias spectrum is then the whole story; relate its decay to standard notions of dataset complexity.
- *PAC learning and entropy.* Relate normalized weight profiles $\overline{W}^k_{\mathcal{D}}$ to PAC-learnability on-distribution, and to entropy-type quantities (an analogue of Hartley entropy for datasets).

Acknowledgments

AI usage: the author would like to acknowledge the use of language models (Gemini and Claude) in drafting and revising this document; all stated identities and theorem constants were additionally verified numerically on random and structured instances.

Bibliography

[1] Ryan O'Donnell. 2021. Analysis of boolean functions. *arXiv preprint arXiv:2105.10386* (2021).

APPENDIX A Appendix

TODO: Deferred proofs and the general finite-abelian-group statements, as they accumulate.